A frozen RNA language model matches hand-crafted features for terminator strength prediction

Siwei Qian

Abstract

Intrinsic termination in bacteria is driven by the nascent transcript alone, and termination efficiency varies widely between terminators in a sequence-dependent way. Efforts to predict termination strength from sequence have so far relied on features drawn from current mechanistic knowledge, which caps what such models can learn. Here I asked whether representations from a pre-trained RNA language model carry information that these features miss. Using the 582 terminators of Chen et al. 2013, I trained ridge regression and XGBoost heads on embeddings from a frozen RNA-FM model. Embeddings from every layer tested outperformed one-hot encoded sequence, and with either head they matched the 16 hand-crafted features of Kundlatsch et al. 2026, which need XGBoost to reach the same level. Adding 19 nt downstream of the termination site improved prediction for one-hot encoding and for some embedding layers. Pre-trained RNA representations are therefore a workable alternative to hand-crafted features for predicting intrinsic terminator strength from sequence.

Introduction

Transcription is a fundamental process of gene expression across all domains of life. Once initiated, transcription must eventually terminate, either at the end of an operon or prematurely within it, in response to specific molecular signals. In bacteria, one form of termination is intrinsic termination, which relies solely on the nascent transcript rather than protein factors[1–3]. The RNA sequence that triggers termination, known as the terminator, typically consists of a stem-loop hairpin followed by a U-rich tract that base pairs with the DNA template. Termination efficiency (TE) is the fraction of RNA transcripts that RNA polymerase (RNAP) releases at a given termination site (Figure 1A). TE is highly variable between terminators and depends on sequence, which allow fine-tuning of gene expression. The current model describes termination as a kinetic competition between elongation (incorporation of the next NTP) and release (Figure 1B)[4,5].

Ever since intrinsic termination was discovered, many efforts have been made to predict it from nucleic acid sequences[4,6–12]. Being able to do so allows discovery of new terminators, especially the weak intragenic ones, and helps in understanding gene regulation across diverse bacterial species. Strength prediction also enables the design of terminators with a range of sequences and strengths, which is a need in synthetic biology and genetic engineering[8].

Over the past several decades, those predictions have been built on mechanistic knowledge of intrinsic termination, including sequence features, thermodynamic parameters, and stochastic modeling[4,7,8,10]. None has reached sufficient accuracy. More recently, machine learning combined with high-throughput in vivo and in vitro measurements has been used to predict termination strength from sequence without assuming a mechanism in the model design[9,11–13]. These studies curate hand-crafted feature sets based on existing knowledge of intrinsic termination and feed them to models such as XGBoost, MLP, or SVM for supervised training. Kundlatsch et al. 2026[11] and Li et al 2025[12] trained on the 582 *E. coli* intrinsic terminator in vivo dual fluorescence reporter dataset from Chen et al 2013[8], and Battaglia et al. 2026[9] used in vitro term-seq data from a library of *B. subtilis* and *E. coli* terminators. Reported mean test R^2 values are in the range of 0.6 to 0.7, but they are not directly comparable, since the studies differ in a variety of aspects, such as the dataset, feature extraction, the scale the target was transformed to, and how the data was split.

Both kinds of approaches are bound by what is currently known about the mechanism, because that knowledge is the input. The mechanistic algorithms are built on it, and hand-crafted features encode it (e.g. number of uridines in the U tract, size of loop, stability of hairpin)[9,11]. That knowledge is incomplete. Multiple features beyond the classic hairpin and U-tract have since been found to change termination, and the intermediates in the mechanism are still debated, so different assumptions lead to different prediction algorithms[1,9,14–16]. It remains unclear whether a different representation of the sequence would carry information that hand-crafted features miss.

Recently, the development of RNA language models has opened opportunities to improve RNA structure and function prediction[17–19]. These models are trained on large numbers of unannotated RNA sequences and can then be fine-tuned for a variety of regression, classification, and structure prediction tasks, such as predicting the family an RNA belongs to. Their performance has been attributed to the learned embeddings capturing underlying structural and functional information about RNA. I therefore asked whether the embeddings from such a model carry information about termination strength that hand-crafted features do not. I used RNA-FM, an RNA language model trained on 23.7 million

non-coding RNA sequences from diverse species, which has been shown to predict non-coding RNA (ncRNA) structures and functions[18]. Note that ncRNA is a broad category; for instance, a tRNA and an intrinsic terminator are very different RNA complexes, so whether the pretrained embeddings transfer to terminators is a hypothesis rather than a given.

Here, I trained ridge regression and XGBoost heads on embeddings from several layers of a frozen RNA-FM model, using the 582 intrinsic terminators in Chen et al. 2013. I found that embeddings from every layer outperform one-hot encoded sequence. With either a ridge or an XGBoost head, they also match the 16 hand-crafted features from Kundlatsch et al. 2026, which need XGBoost to reach the same level. Adding 19 nt downstream of the termination site improves prediction for one-hot encoding and for some embedding layers. Pre-trained RNA representations are therefore a workable alternative to hand-crafted features for predicting intrinsic terminator strength from sequence.

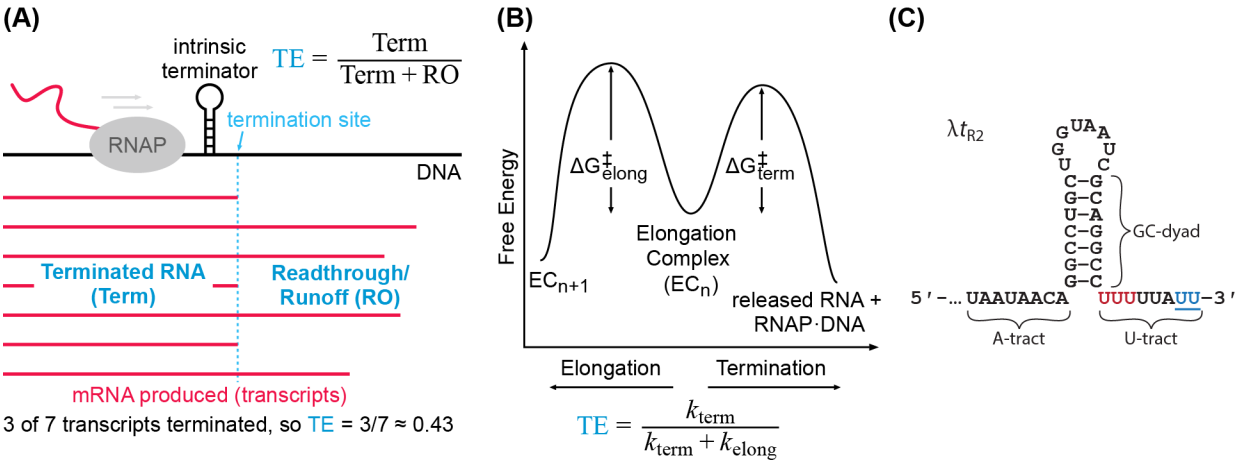


Figure 1. Intrinsic termination and how its efficiency is defined. (A) Termination efficiency (TE) is the fraction of RNA transcript released by RNAP right at a given intrinsic terminator's termination site. In this example 3 of 7 transcripts terminate at the site and 4 read through, so TE = 3/7 ≈ 0.43. (B) Termination is a kinetic competition between adding the next nucleotide and releasing the transcript[1,4]. The two definitions in (A) and (B) describe the same quantity. (C) An example intrinsic terminator λtR2 from *E. coli*, showing the GC-rich hairpin, the U rich RNA-DNA hybrid (U tract) that follows it, and the upstream A-tract. Adapted from Ray-Soni et al. 2016[1].

Method

SOURCE DATA: The data is from Chen et al. 2013, an in vivo dual fluorescence reporter (GFP-intrinsic terminator-RFP) dataset covering 582 natural and synthetic intrinsic terminators[8]. Termination strength (Ts) is reported as (GFP/RFP) / (GFP_0/RFP_0), so

termination efficiency (TE) is 1 − 1/Ts. Each Ts value is the average from at least 3 replicates measured on different days. I originally planned to train on logit(TE), which has a direct correspondence to the transition state activation energy barrier, since TE = k_term / (k_term + k_elong). However, 15% of the terminators in the dataset (90 of 582) have Ts < 1, which makes TE negative. Those values are meaningless in the termination context, since they would mean a lower termination to readthrough ratio than the no-terminator case. To stay aligned with previous modeling studies on this dataset I kept all the data and trained on both Ts and ln(Ts). Ts was the training target in the two previous studies[11,12].

ln(Ts) was selected for several reasons. The measurement error is multiplicative rather than additive in Ts (Figure 2A). Ts spans roughly 0 to 300, so the sum of squares is dominated by the top of the range and a model fit on Ts is graded heavily on the strongest terminators. ln(Ts) also preserves the ranking, so nothing about relative strength is lost. Its relationship to the activation energy barrier is ln(Ts) = −ln(1 − TE) = ln(1 + k_term/k_elong). See Figure 2 for more details.

About half of the Chen et al. 2013 dataset comes from mutating a particular group of intrinsic terminators, so those terminators often share hairpin, U-tract, or A-tract sequence. Grouping the dataset by hairpin type gives 387 groups, of which 311 contain a single terminator and the rest contain at least two. Terminators with similar sequences likely have similar termination strength, so a random split that puts them on both sides would leak information from the training set into the test set. I therefore ran both KFold (random splitting) and GroupKFold (splitting so that terminators sharing a hairpin cannot appear in both the training and the test set) from sklearn.

The sequence window runs from 90 nt upstream of the termination site to 19 nt downstream, in order to cover as much of the sequence that could affect intrinsic termination and Ts readout as possible (Figure 3). The upstream bound was chosen from the distribution of terminator lengths in Chen et al. 2013 and the TERMITe atlas (Figure 3)[20]. The longest is 77 nt in Chen et al. 2013 and 80 nt in TERMITe. The two datasets use different terminator structure prediction algorithms, so the two conventions are not directly comparable, but both are fully covered at 80 nt. I used 90 nt, leaving some margin.

The downstream bound follows Battaglia et al. 2026[9], and also Chen et al. 2013's evidence-based conjecture about one terminator pheA-1 they tested, whose Ts was about 8 fold higher than their biophysics model predicted. They reasoned that pheA-1's hairpin can form a pseudoknot with downstream RNA when the transcript fails to terminate, and that this secondary structure reduces translation of RFP and so increases the observed Ts. Downstream DNA sequence is also known to affect the speed of termination. Because the effect of the downstream region is unclear, I trained on two windows for comparison:

upstream only (90 nt), and upstream plus downstream (109 nt) (Figure 4).

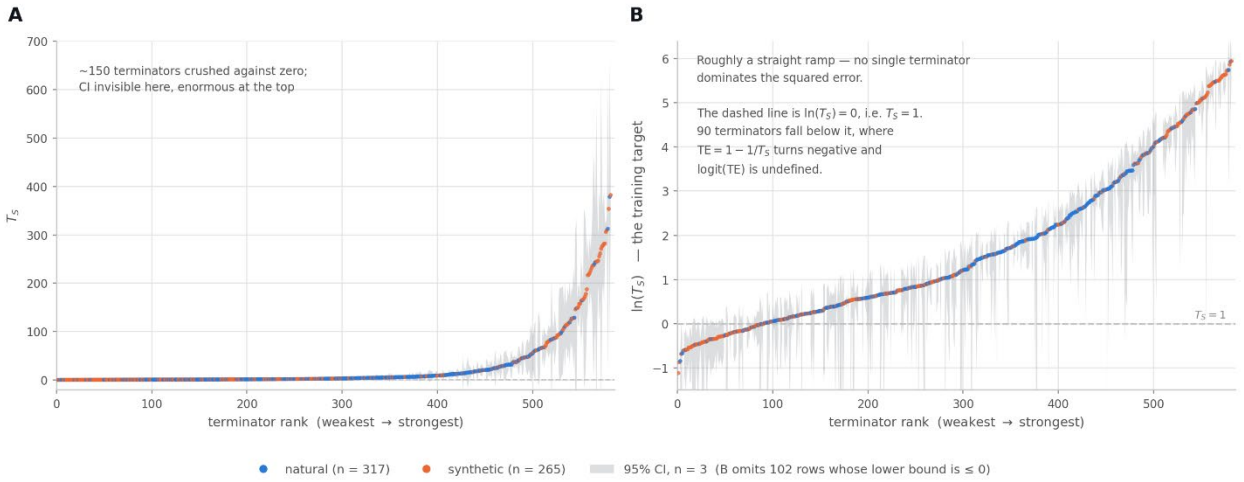


Figure 2. Why ln(Ts) rather than Ts. The 582 Chen 2013 terminators ranked weakest to strongest, same data on both panels. (A) Linear axis. Most terminators have Ts of order 1 to 10 and cannot be told apart on this axis, while the sum of squares is dominated by the few dozen strongest. (B) ln(Ts), the training target. The ramp is close to linear and no single terminator dominates the squared error. The confidence band is close to constant width on this scale, meaning the measurement error is a roughly constant percentage of Ts. The dashed line marks ln(Ts) = 0; the 90 terminators below it have Ts < 1, where TE = 1 − 1/Ts is negative and logit(TE) is undefined. Grey bands are 95% confidence intervals from the per-construct standard deviation, computed as $t(n-1) \times s/\sqrt{n}$ with n = 3 replicates as reported. Lower bounds are floored at Ts/10, so the widest bands reflect that floor rather than the measured uncertainty. Panel B omits 102 rows whose lower bound is ≤ 0.

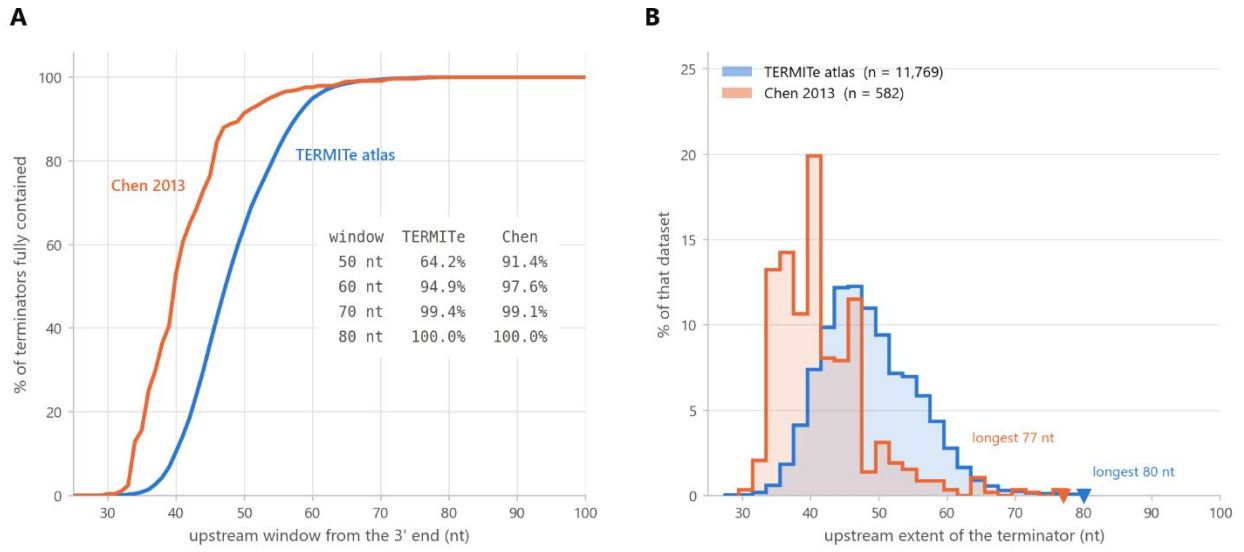


Figure 3. Choice of the upstream window. (A) Fraction of terminators fully contained

within an upstream window of a given length, measured from the 3' end, for Chen et al 2013 (n = 582) and the TERMITe atlas (n = 11,769)[8,20]. Both datasets reach 100% at 80 nt. (B) Distribution of upstream terminator extent in the two datasets.

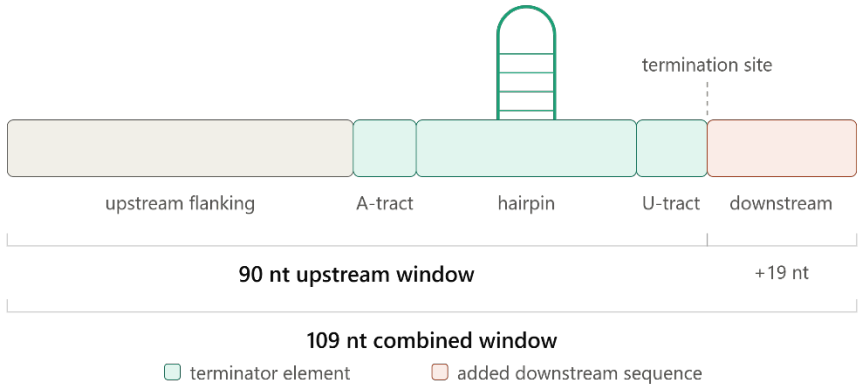


Figure 4. Sequence windows used as model input. Schematic, not to scale. The 90 nt upstream window ends at the termination site; the 109 nt window adds 19 nt downstream.

EMBEDDING EXTRACTION: RNA-FM is a 99M parameter BERT-style transformer that produces a 640-dimensional embedding per nucleotide[18]. The pretrained RNA-FM model was obtained from cuhkaih/rnafm hugging face page and embeddings were extracted following the ml4bio/RNA-FM github repository. Terminator sequences of 90 nt (upstream only) and 109 nt (upstream plus downstream) were given to RNA-FM. The start of the sequence is tokenized as <cls>, the end as <eos>, and every nucleotide in between as its own token. Embeddings were taken from layers 6, 9 and 12 and mean pooled across nucleotide positions, reducing each sequence from (90, 640) or (109, 640), after dropping the <cls> and <eos> embeddings, to a single 640-dimensional vector. The layer 12 <cls> embedding was extracted separately, since pretraining may concentrate sequence-level summary information in that token and mean pooling across all positions would dilute it. As a control, the same sequences were one-hot encoded and flattened to 360 dimensions (90 × 4) or 436 dimensions (109 × 4), giving the heads the sequence alone with nothing from RNA-FM. These are the five input arms compared below.

HAND-CRAFTED FEATURES: The 16 hand-crafted features from Kundlatsch et al. 2026 were regenerated by me from the paper, since neither the feature table nor the training code for their strength model is in their GitHub repository (as of June 2026). Two errors in their described pipeline were fixed: loop %GC counted only G, and the A-tract window direction was reversed. I also parsed the hairpin differently. In their public code the second arm of the stem is built as the reverse complement of the first, so bulges, internal loops and G-U pairs are lost. Here the stem is the paired nucleotides in the dot-bracket structure from Chen et al. 2013, and stem entropy is computed on the actual sequence of both

arms. Hairpin length, state changes and initial GC follow their convention. Reproducing their feature engineering and XGBoost training with these changes gives R^2 0.639 ± 0.147 under their original settings of 5 fold random KFold on Ts.

HEADS: Each input arm was split by 10 fold random KFold and by 10 fold GroupKFold grouped on hairpin type. Seed = 42 was used for KFold and for XGBoost. Under KFold, 39% of test terminators share a hairpin with at least one training terminator. Each fold was then fitted with ridge regression (sklearn.linear_model.RidgeCV, alphas = np.logspace(-3, 6, 40)). Ridge regression was chosen because the embedding dimension is much larger than the number of samples. All ridge arms used the same pipeline of StandardScaler followed by RidgeCV, constructed fresh for each fold and fitted on the training fold only.

For XGBoost, the hyperparameters from Kundlatsch et al. 2026 were used (n_estimators=4000, learning_rate=0.001, max_depth=5, min_child_weight=3, subsample=0.65, random_state=42)[11]. XGBoost was run without scaling, since tree splits do not depend on feature scale. Kundlatsch et al. 2026 used 5 fold random KFold and a Ts target, while this study uses 10 fold GroupKFold and ln(Ts), so the numbers reported here are not directly comparable to their published value.

MODEL EVALUATION: Prediction accuracy was evaluated as R^2 (sklearn.metrics.r2_score) computed per fold and reported as mean ± sd across folds. The rank order correlation between predicted and measured values is reported as the Spearman rank-order correlation coefficient (scipy.stats.spearmanr). Paired differences in R^2 between arms were compared across the 10 cross-validation folds using the Wilcoxon signed-rank test (scipy.stats.wilcoxon, default settings, two-sided). The number of folds in which one arm scored higher is reported alongside the p value.

Package versions: scikit-learn 1.9.0, xgboost 3.2, torch==2.13.0, rna-fm==0.2.2, scipy==1.17.1. Code is available at https://github.com/lingyinfuwafuwa/TE_model

Results

Choosing the regression target and the cross validation split

Before comparing input types, I tested two settings that apply to every comparison below: whether to train on Ts or ln(Ts), and whether to split by random KFold or by GroupKFold on hairpin type.

ln(Ts) collectively produced higher mean Spearman rho than Ts under both splits (0.781 against 0.584 for KFold, 0.747 against 0.536 for GroupKFold, L12 mean pooling). Rho is

rank based, so it can be compared across the two targets while R^2 cannot. This means training on ln(Ts) better preserves the ranking of terminator strength.

Worth noting that R^2 under Ts is also unstable once hairpins are grouped. In Ts | GroupKFold the R^2 standard deviation across folds is larger than the mean itself for every arm (0.4534 against a mean of 0.2635 for L12 mean pooling). Under ln(Ts) in the same split the standard deviations stay between 0.11 and 0.17.

KFold collectively shows higher R^2 than GroupKFold for every arm (0.691 against 0.595 for L12 mean pooling, 0.586 against 0.467 for one-hot). Terminators sharing a hairpin are likely to have similar strength, so when they fall on both sides of the split the test score is inflated. The gap of about 0.10 is the size of that effect in this dataset.

Therefore ln(Ts) and GroupKFold were used for all comparisons below.

Table 1, R^2 (mean ± sd across 10 folds):

arm	ln(Ts) KFold	ln(Ts) GroupKFold	Ts KFold	Ts GroupKFold
L12 mean	0.691 ± 0.042	0.595 ± 0.147	0.574 ± 0.082	0.264 ± 0.453
L12 <cls>	0.674 ± 0.039	0.560 ± 0.165	0.547 ± 0.085	0.219 ± 0.489
L9 mean	0.678 ± 0.039	0.578 ± 0.137	0.545 ± 0.120	0.293 ± 0.343
L6 mean	0.668 ± 0.047	0.573 ± 0.132	0.485 ± 0.087	0.219 ± 0.359
one-hot	0.586 ± 0.056	0.467 ± 0.108	0.438 ± 0.069	0.167 ± 0.322

Table 2, Spearman rho (mean ± sd across 10 folds):

arm	ln(Ts) KFold	ln(Ts) GroupKFold	Ts KFold	Ts GroupKFold
L12 mean	0.781 ± 0.051	0.747 ± 0.093	0.584 ± 0.137	0.536 ± 0.188
L12 <cls>	0.770 ± 0.052	0.734 ± 0.096	0.617 ± 0.091	0.568 ± 0.161
L9 mean	0.769 ± 0.055	0.742 ± 0.089	0.587 ± 0.133	0.542 ± 0.169
L6 mean	0.765 ± 0.054	0.737 ± 0.094	0.632 ± 0.111	0.571 ± 0.194
one-hot	0.722 ± 0.064	0.692 ± 0.075	0.619 ± 0.093	0.582 ± 0.125

RNA-FM embeddings outperform raw-sequence input on intrinsic terminator strength prediction

To test whether RNA-FM representations carry information about intrinsic terminator strength, I extracted embeddings for the 90 nt upstream of the termination site for all terminators in Chen et al. 2013 and used flattened one-hot encoding as a control (see method for details). All five arms used the same RidgeCV head and the same GroupKFold splits, so each fold contains the same terminators across arms.

All four RNA-FM arms outperformed one-hot encoding. Layer 12 mean pooling gave the highest R^2, 0.595 ± 0.147, against 0.467 ± 0.108 for one-hot, a paired difference of 0.128 [wins 8/10, p = 0.010]. Spearman rho gave the same ordering, 0.747 against 0.692. Differences among the four embedding arms were small relative to fold-to-fold variation, so neither the choice of layer nor the choice of pooling readout is resolved at this sample size; layer 12 mean pooling was carried forward as the best performing arm.

Table 3. RNA-FM embeddings compared with one-hot encoded sequences, 90 nt upstream window. Mean ± sd across 10 folds. ΔR^2 is each arm minus one-hot; p is from Wilcoxon signed-rank on the per-fold differences.

arm	R^2	rho	ΔR^2 vs one-hot	wins	p
L12 mean	0.595 ± 0.147	0.747 ± 0.093	+0.128	8/10	0.010
L12 <cls>	0.560 ± 0.165	0.734 ± 0.096	+0.093	8/10	0.049
L9 mean	0.578 ± 0.137	0.742 ± 0.089	+0.111	9/10	0.010
L6 mean	0.573 ± 0.132	0.737 ± 0.094	+0.106	9/10	0.010
one-hot	0.467 ± 0.108	0.692 ± 0.075	ref	ref	ref

Adding 19 nt of downstream sequence improves prediction for L6 embeddings and one-hot encoding

Next I tested whether adding downstream sequence can improve terminator prediction. Previous work showed that readthrough RNA downstream of the terminator can affect the Ts readout by affecting translation (see Methods)[8], and downstream DNA can also affect termination efficiency[16]. Under GroupKFold cross-validation and using ln(Ts) as the regression label, each input type was trained on the 90 nt upstream of the termination site and on the same window plus 19 nt downstream (109 nt). Of all the inputs tested, only L6 mean pooling and one-hot encoding showed a significant increase in R^2 after including downstream sequence (+0.061, p = 0.014 and +0.085, p = 0.014). L9 mean and L12 <cls> both increased but not significantly (+0.033, p = 0.375 and +0.041, p = 0.160), and L12 mean was unchanged (−0.006, p = 1.000). Note that one-hot encoding gained the most of all inputs.

Table 4. Effect of including 19 nt downstream of the termination site. GroupKFold, ln(Ts) target, 10 folds.

arm	R^2, 90 nt	R^2, 109 nt	diff	wins	p
L12 mean	0.595 ± 0.147	0.589 ± 0.148	-0.006	5/10	1.000
L12 <cls>	0.560 ± 0.165	0.600 ± 0.131	+0.041	7/10	0.160
L9 mean	0.578 ± 0.137	0.610 ± 0.145	+0.033	6/10	0.375
L6 mean	**0.573 ± 0.132**	**0.634 ± 0.131**	**+0.061**	**9/10**	**0.014**
one-hot	0.467 ± 0.108	0.552 ± 0.117	+0.085	9/10	0.014

RNA-FM embeddings match hand-crafted features without needing a nonlinear model

To compare RNA-FM embeddings with the hand-crafted intrinsic termination features used in previous work, I ran a two-by-two comparison: two input types (RNA-FM L6 mean pooled embeddings at 109 nt, and the 16 hand-crafted features from Kundlatsch et al 2026) by two models (RidgeCV and XGBoost)[11]. The RNA-FM embedding configuration is the best performing one from the previous section. All four cells use GroupKFold, ln(Ts), and the same 10 fold. The hand-crafted features were refit here under this protocol rather than taken from the published number, since Kundlatsch et al 2026 used 5 fold random KFold on Ts and those settings are not comparable to the rest of this report. My reproduction of their original protocol gives R^2 0.639 ± 0.147.

RNA-FM embeddings with a ridge head reach R^2 0.634 ± 0.131, and hand-crafted features with XGBoost reach 0.620 ± 0.121. The paired difference is +0.014, with RNA-FM higher in 6 of 10 folds ($p = 0.492$). When both input types use XGBoost, the two are also indistinguishable: the difference is −0.020, with RNA-FM higher in 4 of 10 folds ($p = 0.492$). The fold standard deviations are around 0.12, much larger than these differences in mean R^2, so this comparison cannot tell which input type is better. On this dataset, the two perform equivalently. The RNA-FM arm here includes 19 nt downstream, which the hand-crafted features do not encode. The conclusion is the same if the 90 nt embedding arm is used instead (L12 mean pooling with ridge, R^2 0.595 ± 0.147).

Note that the two input types respond differently to the model type. Hand-crafted features improve a lot when moving from ridge to XGBoost, from 0.429 ± 0.203 to 0.620 ± 0.121, a difference of +0.191 in 10 of 10 folds ($p = 0.002$). RNA-FM embeddings do not, going from 0.634 ± 0.131 to 0.601 ± 0.096, a difference of −0.034 in 3 of 10 folds ($p = 0.084$).

Table 5. RNA-FM embeddings compared with hand-crafted features across two model classes. GroupKFold, ln(Ts) target, 10 folds, mean ± sd.

input	Ridge	XGBoost
RNA-FM L6 mean, 109 nt	**0.634 ± 0.131**	0.601 ± 0.096
16 hand-crafted features	0.429 ± 0.203	**0.620 ± 0.121**

Table 6. Paired comparisons, same folds throughout. ΔR^2 is arm 1 minus arm 2; wins is the number of folds in which arm 1 scored higher; p is from Wilcoxon signed-rank on the per-fold differences.

arm 1	arm 2	ΔR^2	wins	p
RNA-FM + Ridge	hand-crafted + XGBoost	+0.014	6/10	**0.492**
RNA-FM + XGBoost	hand-crafted + XGBoost	−0.020	4/10	0.492
RNA-FM + Ridge	hand-crafted + Ridge	+0.205	10/10	0.002
hand-crafted + XGBoost	hand-crafted + Ridge	+0.191	10/10	0.002
RNA-FM + XGBoost	RNA-FM + Ridge	−0.034	3/10	0.084

Discussion

All the RNA-FM embeddings, from every layer and pooling I tested, had higher R^2 than one-hot encoding under both types of cross-validation splits. Since all arms see the same sequence, the gain comes from the form the information is in, not from extra information: with a linear head, the RNA-FM embeddings make some information important for predicting intrinsic termination strength accessible, while the one-hot encoding does not. Worth noting that one-hot alone still reaches $R^2 = 0.467$ under GroupKFold, so the margin is measured against a baseline that is not weak. I did not test one-hot with XGBoost, so whether a nonlinear model could close this gap is not known.

Predictions from RNA-FM embeddings match the performance of the hand-crafted features from Kundlatsch et al. 2026[11] with XGBoost, meaning that RNA-FM embeddings can reach the same level as a recent published approach for predicting termination strength. The hand-crafted features need a nonlinear model to reach a similar R^2, while the embedding performs about the same with both model classes and does not improve with XGBoost. So, the two inputs seem to carry a similar amount of useful information, and

what differs is whether a linear model can use it directly. Note that the XGBoost hyperparameters came from Kundlatsch et al. 2026 and were optimized for the 16 hand-crafted features (see method), and I did not retune them for the 640-dimensional embedding. For the comparison between the two input types this is conservative, since the settings favor the hand-crafted features. For the embedding not improving with XGBoost, it is a real limitation: this may partly reflect settings that were not chosen for the embedding.

Adding 19 nt of downstream sequence improved prediction for L6 embeddings and one-hot, and one-hot gained the most. So the downstream signal can be picked up from raw sequence by a linear model and does not depend on pre-training. This experiment cannot tell why the downstream sequence matters. It may change termination itself[16], or it may change the measured Ts after transcription, as in the pheA-1 case (see Methods)[8], where the hairpin may form a pseudoknot with readthrough RNA and lower downstream RFP translation. L12 mean pooling was the only arm that did not change with the downstream window. Two possible reasons: L12 may already encode what the downstream window adds, or L12 at 90 nt may already be close to the ceiling of this dataset. This experiment cannot tell these apart.

Leakage from shared hairpins is a real problem in this dataset. At the same time, GroupKFold may be too strict. Mutants that share a hairpin can still differ a lot in termination strength, so grouping by hairpin may also block signal that the model could genuinely learn. Both splits are reported for this reason. A better grounded way to group the data would help future work on strength models.

Worth noting what this model is actually predicting. Ts is a GFP/RFP fluorescence ratio measured *in vivo*, not a direct measurement of intrinsic termination. Processes that happen after the RNA is made, such as mRNA processing, translation and protein degradation, also contribute to it. Some of them, such as mRNA degradation has been shown to make a large difference between in vivo and in vitro termination efficiencies on the same intrinsic terminator[10,21]. So every in vivo strength model, mine and the earlier ones, is predicting more than intrinsic termination. This is also a likely reason 90 terminators have Ts < 1. For understanding the mechanism this is worse than using in vitro data. For engineering it may be better, since it may predict how a specific terminator tunes gene expression in vivo more accurately.

There are a few limitations. First, the fold standard deviations are around 0.13, so differences in mean R^2 much smaller than that are not resolvable here. On top of that, the best configuration was chosen by comparing scores on the same folds its R^2 is reported on, so that number is an optimistic estimate of how it would do on new terminators. Second, the dataset is small and not evenly spread. About half of it comes from mutating a small

set of natural terminators, so the sequence space it covers is limited. Third, I only tested one RNA foundation model, and other current models may work as well or better.

As a rough check, I used the per-construct replicate variation reported with the dataset to estimate the highest R^2 any model could reach on these labels[22]. The estimate came out well above what the models here reach, which suggests replicate noise is not the main limit. The real ceiling is probably lower than this estimate, because the replicates do not capture all of the measurement error. The paper does not say whether replicates came from separate clones, and for about half the library the no-terminator control was not run on the same day (and the global control average was used instead). The ceiling has not been carefully validated, so it is not reported here as a number.

Since replicate noise does not look like the limit, there is probably still room to improve termination strength prediction on this kind of data. One next step I am considering is to continue pre-training the model on more than 10,000 unlabeled intrinsic terminator sequences from diverse species, available in public datasets[20]. Other labelled datasets could also be used, such as two in vitro datasets from existing studies[9,23]. Given the power and noise issues above, a larger and cleaner labelled dataset may matter more than any change in method. On a more accurate model, interpretability analysis could be used to extract what the model has learnt about intrinsic termination and whether any of it is new, from attention maps of each layer to methods that pull out internal features. Beyond that, I am interested in modelling intrinsic termination as a continuous-time Markov chain from its mechanism, which is another way to predict termination efficiency. Predictions from either model could then be tested with biochemical assays.

References

1. Ray-Soni, A., Bellecourt, M. J. & Landick, R. Mechanisms of Bacterial Transcription Termination: All Good Things Must End. *Annual Review of Biochemistry* **85**, 319–347 (2016).
2. Peters, J. M., Vangeloff, A. D. & Landick, R. Bacterial Transcription Terminators: The RNA 3'-End Chronicles. *Journal of Molecular Biology* **412**, 793–813 (2011).
3. Julia Dierksheide, K., Battaglia, R. A. & Li, G.-W. How do bacteria tune transcription termination efficiency? *Current Opinion in Microbiology* **82**, 102557 (2024).
4. von Hippel, P. H. & Yager, T. D. Transcript elongation and termination are competitive kinetic processes. *Proceedings of the National Academy of Sciences* **88**, 2307–2311 (1991).
5. Ray-Soni, A., Mooney, R. A. & Landick, R. Trigger loop dynamics can explain stimulation of intrinsic termination by bacterial RNA polymerase without terminator hairpin contact. *Proc. Natl. Acad. Sci. U.S.A.* **114**, (2017).

6. Kingsford, C. L., Ayanbule, K. & Salzberg, S. L. Rapid, accurate, computational discovery of Rho-independent transcription terminators illuminates their relationship to DNA uptake. *Genome Biology* **8**, R22 (2007).
7. Td, Y. & Ph, von H. A thermodynamic analysis of RNA transcript elongation and termination in Escherichia coli. *Biochemistry* **30**, (1991).
8. Chen, Y.-J. *et al.* Characterization of 582 natural and synthetic terminators and quantification of their design constraints. *Nat Methods* **10**, 659–664 (2013).
9. Battaglia, R. A., Ermiş, E., Omollo, E. O. & Li, G.-W. The defining features of intrinsic transcription terminators. 2026.05.29.728758 Preprint at https://doi.org/10.64898/2026.05.29.728758 (2026).
10. Cambray, G. *et al.* Measurement and modeling of intrinsic transcription terminators. *Nucleic Acids Res* **41**, 5139–5148 (2013).
11. Kundlatsch, G. E. *et al.* Machine-learning-driven prediction and design of intrinsic transcription terminators. 2026.01.22.701115 Preprint at https://doi.org/10.64898/2026.01.22.701115 (2026).
12. Li, J., Wu, L.-F., Liu, K. & Ma, B.-G. Intelligent Design of Escherichia coli Terminators by Coupling Prediction and Generation Models. *ACS Synth. Biol.* **14**, 3744–3752 (2025).
13. Koster, C. C. *et al.* Systematic engineering and machine learning analysis of intrinsic terminators reveal crucial nucleotides directly upstream of the terminator hairpin. 2026.07.06.736697 Preprint at https://doi.org/10.64898/2026.07.06.736697 (2026).
14. Roberts, J. W. Mechanisms of Bacterial Transcription Termination. *Journal of Molecular Biology* **431**, 4030–4039 (2019).
15. You, L. *et al.* Structural basis for intrinsic transcription termination. *Nature* **613**, 783–789 (2023).
16. Larson, M. H., Greenleaf, W. J., Landick, R. & Block, S. M. Applied Force Reveals Mechanistic and Energetic Details of Transcription Termination. *Cell* **132**, 971–982 (2008).
17. Yu, H. & Ding, Y. AI foundation models for RNA biology. *RNA Biology* **23**, 1–11 (2026).
18. Chen, J. *et al.* Interpretable RNA Foundation Model from Unannotated Data for Highly Accurate RNA Structure and Function Predictions. Preprint at https://doi.org/10.1101/2022.08.06.503062 (2022).
19. Xu, Y., Pai, N. N. & Wayment-Steele, H. K. Disentangling RNA evolution and thermodynamics in genomic language models. 2026.05.28.728275 Preprint at https://doi.org/10.64898/2026.05.28.728275 (2026).
20. Kosiński, J. G. *et al.* Characterization of bacterial intrinsic transcription terminators identified with TERMITe—a novel method for comprehensive analysis of Term-seq data. *Nucleic Acids Research* **53**, gkaf553 (2025).
21. Lalanne, J.-B. *et al.* Evolutionary Convergence of Pathway-Specific Enzyme Expression Stoichiometry. *Cell* **173**, 749-761.e38 (2018).
22. Li, G. *et al.* Performance of Regression Models as a Function of Experiment Noise. *Bioinform Biol Insights* **15**, 11779322211020315 (2021).
23. Froom, R. *et al.* Cell-free genomics reveals fundamental regulatory principles of the *Mycobacterium tuberculosis* transcription cycle. *Molecular Cell* **86**, 656-673.e12 (2026).